

light. The electron beam constantly scans the screen, starting from the top left corner and ending at the bottom right corner, refreshing one row at a time.

Cathode Ray Tube technology is the predominant—though dated—technology, still powering most monitors. The high voltage needed to power up the display and the space the monitors occupy, in addition to the eye-strain caused by the flickering of the scanning electron beam, are the reasons for looking for alternatives. The ability to display at any resolution without loss of clarity puts them at an advantage over LCD screens.

4.1.8 Dead Pixels

Dead pixels refer to unresponsive elements in a digital display. Since they can no longer control the flow of light from the back panel, these pixels get displayed as spots on the screen that do not change their colour.

4.1.9 DLP

Digital Light Processing is the technology that drives projectors and rear projection TVs. This technology, a creation of Texas Instruments, uses tiltable microscopic mirrors (micromirrors) to manage the intensity of light available at the screen. The light from a source is directed onto a micromirror, whose tilt influences the intensity of reflected light passing to the screen. The position of the mirror when no reflected light reaches the screen is considered the Off position, and the position when the reflected light is available in full intensity is the On position. Each micromirror represents a pixel in the final image. The chip containing the micromirror is called the Digital Micromirror Device (DMD). Each mirror is switched On and Off many thousands of times per second, and depending on the intensity of light needed on the screen, the proportion of the On and Off positions is varied. To get a colour image, the light from the source is passed through a colour wheel containing filters for the three colours (Red, Green, and Blue). Due to the switching mechanism of this technology, the display is subject to the flickering effect as seen in CRTs.